

HDOC – Day 25

C Programming Assignment

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Decreasing Patterns using switch statement, nested loops, arrays and parameterless void functions

Problem requirement: Display a menu and use a switch statement to print the decreasing pattern selected by the user. Each pattern is implemented in a separate parameterless function with no return value.

1. Decreasing right-angled triangle

Input the number of rows n and print a decreasing right-angled triangle using asterisks.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. Print '*' exactly i times using an inner loop.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input	Expected output
1	4	* * * * * * * * * *
2	1	*
3	0	Invalid number of rows

2. Decreasing natural-number pattern

In the first row print natural numbers from 1 to n. Reduce the number of values printed by one in each successive row.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. For j from 1 to i, print j.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected Output
1	4	1 2 3 4 1 2 3 1 2 1
2	2	1 2 1
3	-2	Invalid number of rows.

3. Decreasing repeated-number pattern

Begin with n in the first row and decrease the value by one in each successive row. Print each value as many times as its value.

Algorithm

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. Print the value i exactly i times.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected output
1	4	4 4 4 4 3 3 3 2 2 1
2	3	3 3 3 2 2 1
3	0	Invalid number of rows.

4. Reverse-number triangle

In each row, print natural numbers in descending order, beginning with the number of elements in that row and ending with 1. **Algorithm**

1. START
2. Input n.
3. If $n \leq 0$, display an error and return.
4. For i from n down to 1:
5. For j from i down to 1, print j.
6. Move to the next line.
7. STOP

Test Case Table

Test Case	Input n	Expected output
1	4	4 3 2 1 3 2 1 2 1 1
2	2	2 1 1
3	-1	Invalid number of rows

Complete C Program

```
#include <stdio.h> void
decreasingStarTriangle(void); void
decreasingNaturalPattern(void); void
```

```

decreasingRepeatedPattern(void); void
reverseNumberTriangle(void);
int main(void)
{
int choice;
printf("DECREASING PATTERNS MENU\n");
printf("1. Decreasing right-angled triangle\n");
printf("2. Decreasing natural-number pattern\n");
printf("3. Decreasing repeated-number pattern\n");
printf("4. Reverse-number triangle\n"); printf("Enter
your choice: "); scanf("%d", &choice);
switch (choice)
{
case 1:
decreasingStarTriangle(); break;
case 2:
decreasingNaturalPattern();
break; case 3:
decreasingRepeatedPattern();
break; case 4:
reverseNumberTriangle();
break; default:
printf("Invalid choice.\n");
}
return 0;
}
void decreasingStarTriangle(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) { printf("Invalid number of
rows.\n"); return;
}

```

```

for (i = n; i >= 1; i--) {
for (j = 1; j <= i; j++)
printf("* ");
printf("\n");
}
}

void decreasingNaturalPattern(void)
{
int n, i, j;
printf("Enter number of rows: ");
scanf("%d", &n);
if (n <= 0) { printf("Invalid number of
rows.\n"); return;
}
for (i = n; i >= 1; i--) { for
(j = 1; j <= i; j++)
printf("%d ", j);
printf("\n");
}
}

void decreasingRepeatedPattern(void)
{
int n, i, j;
printf("Enter number of rows: "); scanf("%d",
&n);
if (n <= 0) { printf("Invalid number of
rows.\n"); return;
}
for (i = n; i >= 1; i--) { for
(j = 1; j <= i; j++)
printf("%d ", i);
printf("\n");
}
}

void reverseNumberTriangle(void)

```

```
{  
int n, i, j;  
printf("Enter number of rows: ");  
scanf("%d", &n);  
if (n <= 0) { printf("Invalid number of  
rows.\n"); return;  
}  
for (i = n; i >= 1; i--) { for  
(j = i; j >= 1; j--)  
printf("%d ", j);  
printf("\n");  
}  
}
```

Compilation , Execution & Testing

```
(kali㉿kali)-[~]
$ cc hdoc25.c

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 4
* * *
* *
*
*

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 1
*

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter number of rows: 0
Invalid number of rows.
```

```
(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: 2
1 2
1

(kali㉿kali)-[~]
$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter number of rows: -2
Invalid number of rows.
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 3
3 3 3
2 2
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter number of rows: 0
Invalid number of rows.
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: 2
2 1
1
```

```
(kali㉿kali)-[~]
└─$ ./a.out
DECREASING PATTERNS MENU
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 4
Enter number of rows: -1
Invalid number of rows.
```

